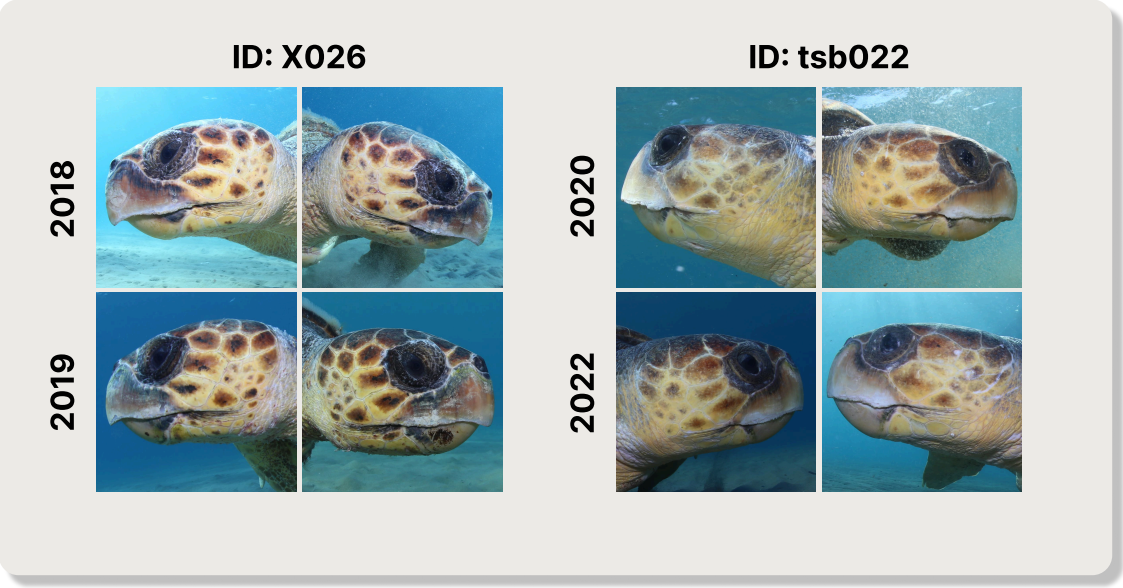
MTH6101, Introduction to Machine Learning: Group Project 2

Building and Comparing Classification Models for Sea Turtle Photo-Identification (Opposite Side Matching)



Background: Sea turtles have geometric facial patterns that are unique to every individual and also stable in time (like human fingerprints!). You can quickly inspect this from the figure above, in which all the photos on the left belong to the same individual (identity name: "X026") and all the photos on the right belong to a different individual (identity name: "tsb022"). Sea Turtle (and in general wildlife) photo-identification is an important tool in wildlife conservation and ecology. Due to the large volume of imaging data, there is a strong interest in automating the process.

Here we are interested in building binary classifiers on pairs of images as follows:

- Do the two images show the *same* sea turtle individual → assign the label "1"
- Do the two images show *different* sea turtle individuals → assign the label "0"

The dataset: You will be working with the "ZakynthosTurtles" which is available to download from https://www.kaggle.com/datasets/wildlifedatasets/zakynthosturtles but it is also available to you in QMplus.
The dataset contains 40 individuals, with four photos per individual: a left- and a right-side photograph taken in one year, and another left- and right-side photograph taken in a different year. As you can see from the figure above, at a given individual, the left and right facial patterns are similar but not exactly the same!

For this project you will be working only with pairs of left-left images of and pairs of right-left images - both for *different* years.

Accompanying feature vectors: Along with every image you will be given an accompanying *feature vector* that belongs to $\mathbb{R}^N$ with $N = 1536$. These feature vectors have been extracted from every image via the *MegaDescriptor* model which is a deep neural network. You will not have to deal with neural networks for this project – you will only make use of these feature vectors. All you need to know is that two feature vectors x, y induce a *similarity* between their corresponding images via their angle (cosine-similarity)

$$s(x, y) = \frac{x \cdot y}{\|x\|\|y\|} \in [-1, 1]$$

→ **Images with high (low) similarity, i.e. close to 1 (far away from 1) are more (less) likely to show the same individual.** The MegaDescriptor model has been trained to achieve this as well as possible.

The task: Your tasks are the following:

(i) Build two threshold-dependent binary classifiers based on the decision rule $s(x, y) > T$: One using pairs of left-left profiles taken in *different* years, and one using pairs of right-left profiles taken in *different* years. There is no training stage; evaluate each classifier using all eligible image pairs.

(ii) Compare the two classifiers. Use ROC curves and AUC. Which one is better? Discuss why could that be? Suggest appropriate values for T and report appropriate metrics. How would you balance sensitivity and specificity in this application? Which of the two might be more important, and why? Why might classification accuracy alone be a misleading performance measure for this dataset?

(iii) Test an LLM of your choice as an alternative classifier. Select a subset of the right-left, different-year image pairs and ask the LLM whether each pair shows the same individual. To ensure that the comparison is meaningful:

- Randomly sample a fixed number of pairs, (e.g. preferably at least 20) from each experiment;
- preserve approximately the same proportion of positive and negative pairs as in the complete dataset;
- Use exactly the same prompt for every pair;
- Force a binary "same"/"different" answer;
- Do not reveal the identity labels to the LLM.

(iv) How does the LLM perform in comparison to your initial models?

(v) Using only the right-profile images, construct a pairwise distance matrix from their feature vectors based on the distance

$$d(x, y) = 1 - s(x, y).$$

Apply agglomerative hierarchical clustering using this distance matrix. Choose and justify an appropriate linkage method. Plot the resulting dendrogram and apply appropriate cuts to obtain clusters. Compare the resulting clusters with the known sea turtle identities and discuss your findings.